Explainable AI Clinical Decision Support System

Clinical Prediction Report

Generated On: 11-08-2026 23:21:09

Patient Information

Patient ID	PAT007
Patient Name	lingesh
Doctor	dhanush
Hospital	CMC
Age	55
Gender	Male
Chest Pain Type	Atypical Angina
Resting Blood Pressure	140 mmHg
Cholesterol	240 mg/dL
Fasting Blood Sugar	No
Rest ECG	ST-T Wave Abnormality
Maximum Heart Rate	150
Exercise Angina	No
ST Depression	1.2
ST Segment Slope	Flat
Major Vessels	0
Thalassemia	Reversible Defect

Prediction Result

Prediction	No Heart Disease
Probability	49.21 %
Risk Level	Low

Clinical Recommendation

No significant indication of heart disease was detected. Maintain a healthy lifestyle, exercise regularly, and undergo routine medical check-ups.

Top Influencing Features (SHAP)

Feature	Impact
Major Vessels	-0.8932
Thalassemia	0.8684
Chest Pain Type	-0.6644
Sex	0.4785
Exercise Induced Angina	-0.2864

LIME Explanation

Feature	Impact
cp <= -0.16	-0.1591
ca <= -0.69	-0.1535
-0.88 < thal <= 1.18	0.111
exang <= -0.70	-0.0761
-1.46 < sex <= 0.68	0.0694

Generated by Explainable AI Clinical Decision Support System

Machine Learning • SHAP • LIME • Flask • React

Doctor Signature: _____

Patient Signature: _____